Dazaet Galicia-Badillo

dazaet.galiciab@outlook.com · (+52) 56 5153 4560 · Mexico City, Mexico

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Dr Daniel Lee
Department of Chemical Engineering, School of Engineering
The University of Manchester
Manchester M13 9PL, United Kingdom

Re: Research Associate in Solid-State NMR Spectroscopy, ref. SAE-032399

Dear Dr Lee,

I am writing to apply for the position of Research Associate in Solid-State NMR Spectroscopy (ref. SAE-032399) within the EPSRC Programme Grant on Molecular Solid Solutions. I completed my PhD in Chemical Sciences at the Institute of Chemistry, UNAM (Mexico), in 2026 under the supervision of Prof. Braulio Rodríguez-Molina, and the degree was awarded with honours.

My doctoral research used solid-state NMR to measure molecular motion in organic crystals and metal-organic frameworks and to connect that motion with crystal structure and macroscopic behaviour. I acquired variable-temperature ^{2}H and ^{13}C solid-state NMR data and analysed them by lineshape simulation and relaxation analysis to obtain rotational rates and activation energies. In a pyrrolo[3,2-b]pyrrole solvate that jumps on heating (Mater. Chem. Front., 2026), I combined ^{2}H NMR with variable-temperature single-crystal diffraction to separate the molecular dynamics from the anisotropic thermal expansion that drives the jump. In a zirconium MOF (ACS Mater. Lett., 2026), I used the same combination to follow how a photoswitch and guest uptake change the motion of a molecular rotor. Both projects were carried out with Dr Martin Dračínský and Dr Jan Blahut at IOCB Prague, and working with them taught me to use NMR and diffraction data as checks on each other.

Several of the desirable criteria describe work I do routinely. I characterise molecular crystals, cocrystals and solvates by SCXRD, PXRD, DSC, TGA and hot-stage microscopy. I run DFT in Gaussian16 and periodic DFT in CP2K, use LAMMPS for molecular dynamics, and wrote two successful proposals for national supercomputing time. Multicomponent molecular crystals are already part of my work: I contributed to a study in which the salient temperature of cocrystals was tuned through the composition of the crystallisation solvent mixture (Angew. Chem. Int. Ed., 2026). Diffraction describes the average structure of a molecular solid solution, while solid-state NMR can show how each component is distributed and how it moves locally. I would very much like to work on that question within MoSS.

My hands-on experience so far is with ^{2}H and ^{13}C. I am keen to extend it to ^{1}H, ^{15}N, ^{19}F and ^{17}O experiments and to DNP-enhanced MAS, and the opportunity to learn these methods in your group is one of the main reasons for this application. During my PhD I learned periodic DFT and molecular dynamics from scratch and now use both in my projects, and I would approach new NMR methods in the same way.

I enjoy working in teams spread across institutions. My 14 publications include collaborations with groups in the Czech Republic, the United States, China and the United Arab Emirates. I co-supervised four bachelor's theses and nine undergraduate research placements, and I received the Best Poster Award at DynaMIC23 in Fribourg. A consortium shared between Durham, Leeds and Manchester, with placements between sites, suits the way I like to work.

Thank you for considering my application. I would be glad to discuss it at interview.

Yours sincerely,

Dazaet Galicia-Badillo, PhD